

```
adityadave@Adityas-MacBook-Air-3 ~ % psql -d tpch-200mb-2
psql (18.4)
Type "help" for help.
```

EXPLAIN (ANALYZE, VERBOSE, BUFFERS)

SET

```

WARNING: Log 1: Planner Input:
DETAIL:      QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE true
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname customer_secure
        :colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone" "
          c_acctbal" "c_mktsegment" "c_comment")
      }
    :rtkind 1
    :subquery
      {QUERY
        :commandType 1
        :querySource 0
        :canSetTag true
        :utilityStmt <>
        :resultRelation 0
        :hasAggs false
        :hasWindowFuncs false
        :hasTargetSRFs false
        :hasSubLinks true
        :hasDistinctOn false
        :hasRecursive false
        :hasModifyingCTE false
        :hasForUpdate false
        :hasRowSecurity false
        :hasGroupRTE false
        :isReturn false
        :cteList <>
        :rtable (
          {RANGETBLENTRY
            :alias
              {ALIAS
                :aliasname c
                :colnames <>
              }
            :eref
              {ALIAS
                :aliasname c
                :colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_ph
                  one" "c_acctbal" "c_mktsegment" "c_comment")
              }
            :rtkind 0
            :reliid 18171
            :inh true
            :relkind r
            :rellockmode 1
            :perminfoindex 1
            :tablesample <>
            :lateral false
            :inFromC1 true
            :securityQuals <>
          }
        )
      }
    :rteperminfos (
      {RTEPERMISSIONINFO
        :reliid 18171
        :inh true
        :requiredPerms 2
        :checkAsUser 10
        :selectedCols (b 8 9 10 11 12 13 14 15)
        :insertedCols (b)
        :updatedCols (b)
      }
    )
  }
  :jointree
    {FROMEXPR
      :fromlist (
        {RANGETBLREF
          :rtindex 1
        }
      )
    }
  )
  :quals
    {BOOLEXPRESS
      :boolop and
      :args (
        {OPEXPRESS
          :opno 1756
          :opfuncid 1720
          :opresulttype 16
          :opretset false
          :opcollid 0
          :inputcollid 0
          :args (
            {VAR
              :varno 1
              :varttno 6
              :vartype 1700
              :vartypmod 983046
              :varcollid 0
              :varnullingrels (b)
              :varlevelsup 0
              :varreturningtype 0
              :varnosyn 1
              :varattnposyn 6
            }
          )
        }
      )
    }
  )
}

```

```

:location -1
}
{FUNCEXPR
:funcid 1740
:funcresulttype 1700
:funcresultset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 0 0 0 0 0 0 0 0 ]
}
)
:location -1
}
}
:location -1
}
{SUBLINK
:subLinkId 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_
comment")
}
:rtekind 0
:relid 18130
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 18136
:inh true
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 18130
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 18136
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
)
)
:quals
{BOOLEXPR
:boolop and
:args (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opresultset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
}
)
}
)
}

```

```

:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
)
:location -1
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
}
:location -1
}
{SCALARARRAYOEXPR
:opno 1054
:opfuncid 1048
:hashfuncid 0
:negfuncid 0
:use0r true
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location -1
}
}
{ARRAYEXPR
:array_typeid 1014
:array_collid 100
:element_typeid 1042
:elements (
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 6
9 ]
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 11 [ 44 0 0 0 65 77 69 82 73 6
7 65 ]
}
}
)
:multidims false
:list_start -1
:list_end -1
:location -1
}
)
:location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>

```

```

        :groupClause <>
        :groupDistinct false
        :groupingSets <>
        :havingQual <>
        :windowClause <>
        :distinctClause <>
        :sortClause <>
        :limitOffset <>
        :limitCount <>
        :limitOption 0
        :rowMarks <>
        :setOperations <>
        :constraintDeps <>
        :withCheckOptions <>
        :stmt_location -1
        :stmt_len -1
    }
    :location -1
}
)
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname c_custkey
:ressortgroupref 0
:resorigtbl 18171
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 1043
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname c_name
:ressortgroupref 0
:resorigtbl 18171
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 1043
:vartypmod 44
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname c_address
:ressortgroupref 0
:resorigtbl 18171
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname c_nationkey
:ressortgroupref 0
:resorigtbl 18171
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 1042
:vartypmod 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname c_phone
:ressortgroupref 0
:resorigtbl 18171
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0

```

```

:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname c_acctbal
:ressortgroupref 0
:resorigtbl 18171
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 1042
:vartypmod 14
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname c_mktsegment
:ressortgroupref 0
:resorigtbl 18171
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 1043
:vartypmod 121
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location -1
}
:resno 8
:resname c_comment
:ressortgroupref 0
:resorigtbl 18171
:resorigcol 8
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier true
:relid 18206
:inh false
:relkind v
:rellockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname orders
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
}
:rtekind 0
:relid 18182
:inh true
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem_secure
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
}
:rtekind 1
:subquery
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname l
:colnames <>
}
}
:eref
{ALIAS
:aliasname l
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber"

```

```

        "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnfl
        ag" "l_linestatus" "l_shipdate" "l_commitdate" "l_receiptdate"
        "l_shipinstruct" "l_shipmode" "l_comment")
    }
}
:rtkind 0
:relid 18194
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
(RTEPERMISSIONINFO
:relid 18194
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
)
)
:jointree
{FROMEXPR
:fromlist (
(RANGETBLREF
:rtindex 1
)
)
:quals
{BOOLEXP
:boolop and
:args (
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rttable (
(RANGETBLENTRY
:alias
{ALIAS
:aliasname o
:colnames <>
}
:eref
{ALIAS
:aliasname o
:colnames ("o_orderkey" "o_custkey" "o_orderstatus"
"o_totalprice" "o_orderdate" "o_orderpriority" "o_
clerk" "o_shippriority" "o_comment")
}
:rtkind 0
:relid 18182
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
(RTEPERMISSIONINFO
:relid 18182
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
(RANGETBLREF
:rtindex 1
)
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:varatype 23
:varitypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 1
:varatype 23
:varitypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
)
:location -1
}
)
{SCALARARRAYOPEXP

```

```

:opno 1054
:opfuncid 1048
:hashfuncid 0
:negfuncid 0
:useOr true
:inputcollid 100
:args (
  {VAR
    :varno 1
    :varattno 6
    :vartype 1042
    :vartypmod 19
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 6
    :location -1
  }
  {ARRAYEXPR
    :array_typeid 1014
    :array_collid 100
    :element_typeid 1042
    :elements (
      {CONST
        :consttype 1042
        :consttypmod -1
        :constcollid 100
        :constlen -1
        :constbyval false
        :constisnull false
        :location -1
        :constvalue 12 [ 48 0 0 0 49 45 85 82 71 6
          9 78 84 ]
      }
      {CONST
        :consttype 1042
        :consttypmod -1
        :constcollid 100
        :constlen -1
        :constbyval false
        :constisnull false
        :location -1
        :constvalue 10 [ 40 0 0 0 50 45 72 73 71 7
          2 ]
      }
    )
    :multidims false
    :list_start -1
    :list_end -1
    :location -1
  }
)
:location -1
)
:location -1
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {CONST
      :consttype 23
      :consttypmod -1
      :constcollid 0
      :constlen 4
      :constbyval true
      :constisnull false
      :location -1
      :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
)
:location -1
)
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
  {QUERY
    :commandType 1
    :querySource 0
    :canSetTag true
    :utilityStmt <>
    :resultRelation 0
    :hasAggs false
    :hasWindowFuncs false
    :hasTargetSRFs false
    :hasSublinks false
    :hasDistinctOn false
    :hasRecursive false
    :hasModifyingCTE false
    :hasForUpdate false
    :hasRowSecurity false
    :hasGroupRTe false
    :isReturn false
    :ctelist <>
    :rtable (
      {RANGETBLENTRY
        :alias
        {ALIAS
          :aliasname ps
          :colnames <>
        }
        :eref
        {ALIAS
          :aliasname ps
          :colnames ("ps_partkey" "ps_suppkey" "ps_availqty"
            "ps_supplycost" "ps_comment")
        }
        :rtekind 0
        :relid 18163
        :inh true
      }
    )
  }
}

```

```

:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 18163
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
)
:location -1
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
)
:location -1
}
{OEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
)
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 0 0 0 0 0 0 0 0 ]
}
)
:location -1
}
)
:location -1
}
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY

```

```

        :expr
        {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
        }
        :resno 1
        :resname ?column?
        :resortgroupref 0
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
    }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname l_orderkey
:resortgroupref 0
:resorigtbl 18194
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname l_partkey
:resortgroupref 0
:resorigtbl 18194
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname l_suppkey
:resortgroupref 0
:resorigtbl 18194
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname l_linenummer
:resortgroupref 0
:resorigtbl 18194
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)

```

```

:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname 1_quantity
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:varatype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname 1_extendedprice
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:varatype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname 1_discount
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:varatype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location -1
}
:resno 8
:resname 1_tax
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:varatype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location -1
}
:resno 9
:resname 1_returnflag
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:varatype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location -1
}
:resno 10
:resname 1_linestatus
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:varatype 1002
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location -1
}
:resno 11
:resname 1_shipdate
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR

```

```

:varno 1
:varattno 12
:vartype 1002
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location -1
}
:resno 12
:resname l_commitdate
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1002
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location -1
}
}
:resno 13
:resname l_receiptdate
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location -1
}
}
:resno 14
:resname l_shipinstruct
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 14
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 15
:vartype 1042
:vartypmod 14
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 15
:location -1
}
}
:resno 15
:resname l_shipmode
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 15
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 16
:vartype 1043
:vartypmod 48
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 16
:location -1
}
}
:resno 16
:resname l_comment
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 16
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier true
:relid 18295
:inh false
:relkind v
:rellockmode 1
:perminfoindex 3
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGEBLENTY
:alias <>
:eref
{ALIAS
:aliasname supplier_secure
:colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
}

```

```

:rtkind 1
:subquery
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_ph
one" "s_acctbal" "s_comment")
}
:rtkind 0
:relid 18153
:inh true
:relkind R
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 18153
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:varstype 1700
:varstypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
)
{FUNCEXP
:funcid 1740
:funcresulttype 1700
:funcresultset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 0 0 0 0 0 0 0 0 ]
}
)
:location -1
}
)
:location -1
}
)
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_

```

```

        comment")
    }
    :rtekind 0
    :relid 18130
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
  {RANGETBENTRY
  :alias
  {ALIAS
  :aliasname r
  :colnames <>
  }
  :eref
  {ALIAS
  :aliasname r
  :colnames ("r_regionkey" "r_name" "r_comment")
  }
  :rtekind 0
  :relid 18136
  :inh true
  :relkind r
  :rellockmode 1
  :perminfoindex 2
  :tablesample <>
  :lateral false
  :inFromCl true
  :securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 18130
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 18136
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
)
:location -1
}
)
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
)
:location -1
}
)
{OPEXP
:opno 1054
:opfuncid 1048
:opresulttype 16

```

```

        :opretset false
        :opcollid 0
        :inputcollid 100
        :args (
            {VAR
                :varno 2
                :varattno 2
                :vartype 1042
                :vartypmod 29
                :varcollid 100
                :varnullingrels (b)
                :varlevelsup 0
                :varreturningtype 0
                :varnosyn 2
                :varattnosyn 2
                :location -1
            }
            {CONST
                :consttype 1042
                :consttypmod -1
                :constcollid 100
                :constlen -1
                :constbyval false
                :constisnull false
                :location -1
                :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
            }
        )
        :location -1
    }
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
    {TARGETENTRY
        :expr
        {CONST
            :consttype 23
            :consttypmod -1
            :constcollid 0
            :constlen 4
            :constbyval true
            :constisnull false
            :location -1
            :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
        }
        :resno 1
        :resname ?column?
        :resortgroupref 0
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
    }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
:location -1
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
    {TARGETENTRY
        :expr
        {VAR
            :varno 1
            :varattno 1
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 1
            :location -1
        }
        :resno 1
        :resname s_suppkey
        :resortgroupref 0
        :resorigtbl 18153
        :resorigcol 1
        :resjunk false
    }
)
{TARGETENTRY
:expr
    {VAR
        :varno 1
        :varattno 2
        :vartype 1042
        :vartypmod 29
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 2
        :location -1
    }
}
:resno 2
:resname s_name
:resortgroupref 0
:resorigtbl 18153
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
    {VAR
        :varno 1
        :varattno 3
        :vartype 1043
        :vartypmod 44
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 3
        :location -1
    }
}

```

```

    }
:resno 3
:resname s_address
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname s_nationkey
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 1042
:vartypmod 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname s_phone
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname s_acctbal
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 1043
:vartypmod 105
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname s_comment
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 7
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier true
:relid 18291
:inh false
:relkind v
:rellockmode 1
:perminfoindex 4
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname nation
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
}
:rtekind 0
:relid 18130
:inh true
:relkind r
:rellockmode 1
:perminfoindex 5
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY

```

```

:alias <>
:eref
{ALIAS
:aliasname region
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 18136
:inh true
:rtekind r
:rellockmode 1
:perminfoindex 6
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTTRY
:alias <>
:eref
{ALIAS
:aliasname *GROUP*
:colnames ("n_name")
}
:rtekind 9
:groupexprs (
{VAR
:varno 5
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 2
:location 51
}
)
:lateral false
:inFromCl false
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 18286
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 18182
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 12)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 18295
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10 13 14)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 18291
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 18130
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 18136
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
{RANGETBLREF
:rtindex 3
}
{RANGETBLREF
:rtindex 4
}
{RANGETBLREF
:rtindex 5
}
{RANGETBLREF
:rtindex 6
}
)
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 259
}
)
}
)
}

```

```

{VAR
:varno 2
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location 271
}
}
:location 269
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 293
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 306
}
}
:location 304
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 3
:location 329
}
{VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location 341
}
}
:location 339
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 363
}
{VAR
:varno 4
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 4
:location 377
}
}
:location 375
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 4
:location 401
}
}

```

```

{VAR
:varno 5
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location 415
}
}
:location 413
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 5
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 3
:location 439
}
{VAR
:varno 6
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 1
:location 453
}
}
}
:location 451
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 6
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 2
:location 477
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 486
:constvalue 8 [ 32 0 0 0 65 83 73 65 ]
}
}
}
:location 484
}
{OPEXPR
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 5
:location 505
}
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 525
:constvalue 4 [ -34 -8 -1 -1 -1 -1 -1 ]
}
}
}
:location 517
}
{OPEXPR
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 5
:location 550
}
{OPEXPR
:opno 1076
:opfuncid 2071
:opresulttype 1114
:opretset false
:opcollid 0

```

```

        :inputcollid 0
      :args (
        {CONST
          :consttype 1082
          :consttypmod -1
          :constcollid 0
          :constlen 4
          :constbyval true
          :constisnull false
          :location 569
          :constvalue 4 [ -34 -8 -1 -1 -1 -1 -1 ]
        }
        {CONST
          :consttype 1186
          :consttypmod 327679
          :constcollid 0
          :constlen 16
          :constbyval false
          :constisnull false
          :location 593
          :constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 0 12 0 0 0 ]
        }
      )
    :location 582
  }
}
:location 562
}
}
:location 289
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 7
      :varattno 1
      :vartype 1042
      :vartypmod 29
      :varcollid 100
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 7
      :varattnosyn 1
      :location -1
    }
    :resno 1
    :resname n_name
    :ressortgrouppref 2
    :resorigtbl 18130
    :resorigcol 2
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {AGGREF
      :aggfnoid 2114
      :aggtype 1700
      :aggcollid 0
      :inputcollid 0
      :aggtranstype 0
      :aggargtypes (o 1700)
      :aggdirectargs <>
      :args (
        {TARGETENTRY
          :expr
          {OEXPR
            :opno 1760
            :opfuncid 1726
            :opresulttype 1700
            :opretset false
            :opcollid 0
            :inputcollid 0
            :args (
              {VAR
                :varno 3
                :varattno 6
                :vartype 1700
                :vartypmod 983046
                :varcollid 0
                :varnullingrels (b)
                :varlevelsup 0
                :varreturningtype 0
                :varnosyn 3
                :varattnosyn 6
                :location 71
              }
              {OEXPR
                :opno 1759
                :opfuncid 1725
                :opresulttype 1700
                :opretset false
                :opcollid 0
                :inputcollid 0
                :args (
                  {FUNCEXP
                    :funcid 1740
                    :funcresulttype 1700
                    :funcset false
                    :funcvariadic false
                    :funcformat 2
                    :funccollid 0
                    :inputcollid 0
                    :args (
                      {CONST
                        :consttype 23
                        :consttypmod -1
                        :constcollid 0
                        :constlen 4
                        :constbyval true
                        :constisnull false
                        :location 90
                        :constvalue 4 [ 1 0 0 0 0 0 0 ]
                      }
                    )
                  :location -1
                }
                {VAR
                  :varno 3
                  :varattno 7
                  :vartype 1700
                  :vartypmod 983046
                  :varcollid 0
                  :varnullingrels (b)
                  :varlevelsup 0
                  :varreturningtype 0
                  :varnosyn 3
                  :varattnosyn 7
                  :location 94
                }
              )
            :location 92
          }
        }
      :location 87
    }
    :resno 1
    :resname <>
    :ressortgrouppref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)

```

```

    )
    :aggorder <>
    :aggdistinct <>
    :aggfilter <>
    :aggstar false
    :aggvariadic false
    :aggkind n
    :agppresorted false
    :agglevelsup 0
    :aggsplit 0
    :aggn -1
    :aggttransno -1
    :location 67
  }
  :resno 2
  :resname revenue
  :resortgroupref 1
  :resorigtbl 0
  :resorigcol 0
  :resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
  {SORTGROUPCLAUSE
   :tleSortGroupRef 2
   :eqop 1054
   :sortop 1058
   :reverse_sort false
   :nulls_first false
   :hashable true
  }
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
  {SORTGROUPCLAUSE
   :tleSortGroupRef 1
   :eqop 1752
   :sortop 1756
   :reverse_sort true
   :nulls_first true
   :hashable true
  }
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

WARNING: max_parallel_hazard_walker: FuncExpr funcid=1740 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=1740 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=1740 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE true
:isReturn false
:cteList <>
:rtable (
  {RANGETBENTRY
   :alias <>
   :eref
   {ALIAS
    :aliasname customer_secure
    :colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone" "
c_acctbal" "c_mktsegment" "c_comment")
   }
  }
:rtekind 1
:subquery
  {QUERY
   :commandType 1
   :querySource 0
   :canSetTag true
   :utilityStmt <>
   :resultRelation 0
   :hasAggs false
   :hasWindowFuncs false
   :hasTargetSRFs false
   :hasSubLinks true
   :hasDistinctOn false
   :hasRecursive false
   :hasModifyingCTE false
   :hasForUpdate false
   :hasRowSecurity false
   :hasGroupRTE false
   :isReturn false
   :cteList <>
   :rtable (
    {RANGETBENTRY
     :alias
     {ALIAS
      :aliasname c
      :colnames <>
     }
    }
   :eref
   {ALIAS
    :aliasname c
    :colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_ph
one" "c_acctbal" "c_mktsegment" "c_comment")
   }
  }
:rtekind 0
:relid 18171
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCI true
:securityQuals <>
}
)
:rteperminfos (
  {RTEPERMISSIONINFO
   :relid 18171
   :inh true
   :requiredPerms 2
   :checkAsUser 10
   :selectedCols (b 8 9 10 11 12 13 14 15)
   :insertedCols (b)
   :updatedCols (b)
  }
)

```

```

)
:jointree
{FROMEXPR
:fromlist (
(RANGETBREF
:rtnindex 1
)
)
:quals
{BOOLEXP
:boolop and
:args (
{OEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:varatype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
)
}{FUNCEXP
:funcid 1740
:funcresulttype 1700
:funcresultset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 0 0 0 0 0 0 0 0 ]
}
)
:location -1
}
)
:location -1
)
}{SUBLINK
:sublinktype 0
:sublinkid 0
:testexpr <>
:opername <>
:subselect
{QUERY
:commandtype 1
:querysource 0
:cansettag true
:utilitystmt <>
:resultrelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctclset <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:reldid 18130
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:reldid 18136
:inh true
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
}
:rteperminfos (
{RTEPERMISSIONINFO
:reldid 18130
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:reldid 18136
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
}

```

```

    }
}
:jointree
{FROMEXPR
  :fromlist (
    {RANGETBLREF
      :rtindex 1
    }
    {RANGETBLREF
      :rtindex 2
    }
  )
}
:quals
{BOOLEXP
  :boolop and
  :args (
    {OPEXP
      :opno 96
      :opfuncid 65
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        {VAR
          :varno 1
          :varattno 1
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 1
          :location -1
        }
        {VAR
          :varno 1
          :varattno 4
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 1
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 4
          :location -1
        }
      )
    }
    :location -1
  )
}
{OPEXP
  :opno 96
  :opfuncid 65
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
    {VAR
      :varno 1
      :varattno 3
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 3
      :location -1
    }
    {VAR
      :varno 2
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 2
      :varattnosyn 1
      :location -1
    }
  )
}
:location -1
}
{SCALARARRAYOPEXP
  :opno 1054
  :opfuncid 1048
  :hashfuncid 0
  :negfuncid 0
  :useOr true
  :inputcollid 100
  :args (
    {VAR
      :varno 2
      :varattno 2
      :vartype 1042
      :vartypmod 29
      :varcollid 100
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 2
      :varattnosyn 2
      :location -1
    }
    {ARRAYEXP
      :array_typeid 1014
      :array_collid 100
      :element_typeid 1042
      :elements (
        {CONST
          :consttype 1042
          :consttypmod -1
          :constcollid 100
          :constlen -1
          :constbyval false
          :constisnull false
          :location -1
          :constvalue 10 [ 40 0 0 0 69 85 82 79 80 6
            9 ]
        }
        {CONST
          :consttype 1042
          :consttypmod -1
          :constcollid 100
          :constlen -1
          :constbyval false
          :constisnull false
          :location -1
          :constvalue 11 [ 44 0 0 0 65 77 69 82 73 6
            7 65 ]
        }
      )
    }
  )
  :multidims false
  :list_start -1
  :list_end -1
  :location -1
}
}
:location -1
}
)

```

```

        :location -1
    }
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
    {TARGETENTRY
    :expr
    {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
)
:override 0
:onConflict <>
:ReturningOldAlias <>
:ReturningNewAlias <>
:ReturningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
:windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location -1
    }
    :resno 1
    :resname c_custkey
    :ressortgroupref 0
    :resorigtbl 18171
    :resorigcol 1
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 2
    :vartype 1043
    :vartypmod 29
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location -1
    }
    :resno 2
    :resname c_name
    :ressortgroupref 0
    :resorigtbl 18171
    :resorigcol 2
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 3
    :vartype 1043
    :vartypmod 44
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
    }
    :resno 3
    :resname c_address
    :ressortgroupref 0
    :resorigtbl 18171
    :resorigcol 3
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 4
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 4
    :location -1
    }
    :resno 4
    :resname c_nationkey
    :ressortgroupref 0
    :resorigtbl 18171
    :resorigcol 4
    :resjunk false
    }
    {TARGETENTRY

```

```

:expr
{VAR
:varno 1
:varattno 5
:vartype 1042
:vartypmod 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname c_phone
:ressortgroupref 0
:resorigtbl 18171
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname c_acctbal
:ressortgroupref 0
:resorigtbl 18171
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 1042
:vartypmod 14
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname c_mktsegment
:ressortgroupref 0
:resorigtbl 18171
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 1043
:vartypmod 121
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location -1
}
:resno 8
:resname c_comment
:ressortgroupref 0
:resorigtbl 18171
:resorigcol 8
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier true
:relid 18286
:inh false
:relkind v
:rellockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname orders
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
}
:rtekind 0
:relid 18182
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem_secure
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
}

```

```

    }
:rtekind 1
:subquery
  {QUERY
    :commandType 1
    :querySource 0
    :canSetTag true
    :utilityStmt <>
    :resultRelation 0
    :hasAggs false
    :hasWindowFuncs false
    :hasTargetSRFs false
    :hasSubLinks true
    :hasDistinctOn false
    :hasRecursive false
    :hasModifyingCTE false
    :hasForUpdate false
    :hasRowSecurity false
    :hasGroupRTE false
    :isReturn false
    :ctelist <>
    :rtable (
      {RANGETBLENTRY
        :alias
        {ALIAS
          :aliasname 1
          :colnames <>
        }
        :eref
        {ALIAS
          :aliasname 1
          :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenummer"
            "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnfl
            ag" "l_linestatus" "l_shipdate" "l_commitdate" "l_receiptdate"
            "l_shipinstruct" "l_shipmode" "l_comment")
        }
      }
    :rtekind 0
    :relid 18194
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCI true
    :securityQuals <>
  )
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 18194
    :inh true
    :requiredPerms 2
    :checkAsUser 10
    :selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
  {FROMEXPR
    :fromlist (
      {RANGETBLREF
        :rtindex 1
      }
    )
  }
:quals
  {BOOLEXP
    :boolop and
    :args (
      {SUBLINK
        :subLinkType 0
        :subLinkId 0
        :testexpr <>
        :operName <>
        :subselect
          {QUERY
            :commandType 1
            :querySource 0
            :canSetTag true
            :utilityStmt <>
            :resultRelation 0
            :hasAggs false
            :hasWindowFuncs false
            :hasTargetSRFs false
            :hasSubLinks false
            :hasDistinctOn false
            :hasRecursive false
            :hasModifyingCTE false
            :hasForUpdate false
            :hasRowSecurity false
            :hasGroupRTE false
            :isReturn false
            :ctelist <>
            :rtable (
              {RANGETBLENTRY
                :alias
                {ALIAS
                  :aliasname 0
                  :colnames <>
                }
                :eref
                {ALIAS
                  :aliasname 0
                  :colnames ("o_orderkey" "o_custkey" "o_orderstatus"
                    "o_totalprice" "o_orderdate" "o_orderpriority" "o_
                    clerk" "o_shippriority" "o_comment")
                }
              }
            :rtekind 0
            :relid 18182
            :inh true
            :relkind r
            :rellockmode 1
            :perminfoindex 1
            :tablesample <>
            :lateral false
            :inFromCI true
            :securityQuals <>
          )
        }
      }
    :rteperminfos (
      {RTEPERMISSIONINFO
        :relid 18182
        :inh true
        :requiredPerms 2
        :checkAsUser 10
        :selectedCols (b 8 13)
        :insertedCols (b)
        :updatedCols (b)
      }
    )
  }
:jointree
  {FROMEXPR
    :fromlist (
      {RANGETBLREF
        :rtindex 1
      }
    )
  }
:quals
  {BOOLEXP
    :boolop and
    :args (
      {OPEXP
        :opno 96
        :opfuncid 65
        :opresulttype 16
        :opretset false
        :opcollid 0
      }
    )
  }

```

```

:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location -1
  }
  {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 1
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location -1
  }
)
:location -1
}
{SCALARARRAYOPEXPR
:opno 1054
:opfuncid 1048
:hashfuncid 0
:negfuncid 0
:useOr true
:inputcollid 100
:args (
  {VAR
    :varno 1
    :varattno 6
    :vartype 1042
    :vartypmod 19
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 6
    :location -1
  }
  {ARRAYEXPR
:array_typeid 1014
:array_collid 100
:element_typeid 1042
:elements (
  {CONST
    :consttype 1042
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location -1
    :constvalue 12 [ 48 0 0 0 49 45 85 82 71 6
9 78 84 ]
  }
  {CONST
    :consttype 1042
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location -1
    :constvalue 10 [ 40 0 0 0 50 45 72 73 71 7
2 ]
  }
)
:multidims false
:list_start -1
:list_end -1
:location -1
}
)
:location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetlist (
  {TARGETENTRY
:expr
  {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
  }
  :resno 1
  :resname ?column?
  :ressortgroupref 0
  :resorigtbl 0
  :resorigcol 0
  :resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
  {QUERY
:commandType 1

```

```

:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
  {RANGETBENTRY
    :alias
      {ALIAS
        :aliasname ps
        :colnames <>
      }
    :eref
      {ALIAS
        :aliasname ps
        :colnames ("ps_partkey" "ps_suppkey" "ps_availqty"
          "ps_supplycost" "ps_comment")
      }
    :rtekind 0
    :relid 18163
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 18163
    :inh true
    :requiredPerms 2
    :checkAsUser 10
    :selectedCols (b 8 9 10)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
  {FROMEXPR
    :fromlist (
      {RANGETBLREF
        :rtindex 1
      }
    )
    :quals
      {BOOLEXP
        :boolop and
        :args (
          {OEXPR
            :opno 96
            :opfuncid 65
            :opresulttype 16
            :opretset false
            :opcollid 0
            :inputcollid 0
            :args (
              {VAR
                :varno 1
                :varattno 1
                :vartype 23
                :vartypmod -1
                :varcollid 0
                :varnullingrels (b)
                :varlevelsup 0
                :varreturningtype 0
                :varnosyn 1
                :varattnosyn 1
                :location -1
              }
              {VAR
                :varno 1
                :varattno 2
                :vartype 23
                :vartypmod -1
                :varcollid 0
                :varnullingrels (b)
                :varlevelsup 1
                :varreturningtype 0
                :varnosyn 1
                :varattnosyn 2
                :location -1
              }
            )
            :location -1
          }
          {OEXPR
            :opno 96
            :opfuncid 65
            :opresulttype 16
            :opretset false
            :opcollid 0
            :inputcollid 0
            :args (
              {VAR
                :varno 1
                :varattno 2
                :vartype 23
                :vartypmod -1
                :varcollid 0
                :varnullingrels (b)
                :varlevelsup 0
                :varreturningtype 0
                :varnosyn 1
                :varattnosyn 2
                :location -1
              }
              {VAR
                :varno 1
                :varattno 3
                :vartype 23
                :vartypmod -1
                :varcollid 0
                :varnullingrels (b)
                :varlevelsup 1
                :varreturningtype 0
                :varnosyn 1
                :varattnosyn 3
                :location -1
              }
            )
            :location -1
          }
          {OEXPR
            :opno 521
            :opfuncid 147
            :opresulttype 16
            :opretset false
            :opcollid 0
            :inputcollid 0
            :args (
              {VAR
                :varno 1
                :varattno 3
              }
            )
          }
        )
      }
    }
  }

```

```

        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 3
        :location -1
    }
    {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 4 [ 0 0 0 0 0 0 0 0 ]
    }
    )
    :location -1
    }
    )
    :location -1
    }
    }
    :mergeActionList <>
    :mergeTargetRelation 0
    :mergeJoinCondition <>
    :targetlist (
    {TARGETENTRY
    :expr
    {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    )
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause <>
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :rowMarks <>
    :setOperations <>
    :constraintDeps <>
    :withCheckOptions <>
    :stmt_location -1
    :stmt_len -1
    }
    :location -1
    }
    )
    :location -1
    }
    }
    :mergeActionList <>
    :mergeTargetRelation 0
    :mergeJoinCondition <>
    :targetlist (
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location -1
    }
    :resno 1
    :resname l_orderkey
    :ressortgroupref 0
    :resorigtbl 18194
    :resorigcol 1
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location -1
    }
    :resno 2
    :resname l_partkey
    :ressortgroupref 0
    :resorigtbl 18194
    :resorigcol 2
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
    }
    :resno 3
    :resname l_supkey
    :ressortgroupref 0
    :resorigtbl 18194
    :resorigcol 3
    :resjunk false

```

```

}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname l_linenumber
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname l_quantity
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname l_extendedprice
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname l_discount
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location -1
}
:resno 8
:resname l_tax
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location -1
}
:resno 9
:resname l_returnflag
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location -1
}
}

```

```

:resno 10
:resname l_linestatus
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:varypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location -1
}
}
:resno 11
:resname l_shipdate
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:varypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location -1
}
}
:resno 12
:resname l_commitdate
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:varypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location -1
}
}
:resno 13
:resname l_receiptdate
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 1042
:varypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location -1
}
}
:resno 14
:resname l_shipinstruct
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 14
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 15
:vartype 1042
:varypmod 14
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 15
:location -1
}
}
:resno 15
:resname l_shipmode
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 15
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 16
:vartype 1043
:varypmod 48
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 16
:location -1
}
}
:resno 16
:resname l_comment
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 16
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>

```

```

:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier true
:relid 18295
:inh false
:relkind v
:rellockmode 1
:perminfoindex 3
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname supplier_secure
:colnames ("s_suppley" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
:rtekind 1
:subquery
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_suppley" "s_name" "s_address" "s_nationkey" "s_ph
one" "s_acctbal" "s_comment")
}
:rtekind 0
:relid 18153
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 18153
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXPR
:boolop and
:args (
{OPEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{FUNCEXPR
:funcid 1740
:funcresulttype 1700
:funcset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 0 0 0 0 0 0 0 0 ]
}
)
:location -1
}
)
:location -1
}
{SUBLINK
:subLinkType 0
:subLinkId 0

```

```

:testexpr <>
:operName <>
:subselect
  (QUERY
    :commandType 1
    :querySource 0
    :canSetTag true
    :utilityStmt <>
    :resultRelation 0
    :hasAggs false
    :hasWindowFuncs false
    :hasTargetSRFs false
    :hasSubLinks false
    :hasDistinctOn false
    :hasRecursive false
    :hasModifyingCTE false
    :hasForUpdate false
    :hasRowSecurity false
    :hasGroupRTE false
    :isReturn false
    :ctelist <>
    :rtable (
      {RANGETBLENTRY
        :alias
        {ALIAS
          :aliasname n
          :colnames <>
        }
        :eref
        {ALIAS
          :aliasname n
          :colnames ("n_nationkey" "n_name" "n_regionkey" "n_
            comment")
        }
        :rtekind 0
        :relid 18130
        :inh true
        :relkind r
        :rellockmode 1
        :perminfoindex 1
        :tablesample <>
        :lateral false
        :inFromCI true
        :securityQuals <>
      }
      {RANGETBLENTRY
        :alias
        {ALIAS
          :aliasname r
          :colnames <>
        }
        :eref
        {ALIAS
          :aliasname r
          :colnames ("r_regionkey" "r_name" "r_comment")
        }
        :rtekind 0
        :relid 18136
        :inh true
        :relkind r
        :rellockmode 1
        :perminfoindex 2
        :tablesample <>
        :lateral false
        :inFromCI true
        :securityQuals <>
      }
    )
    :rteperminfos (
      {RTEPERMISSIONINFO
        :relid 18130
        :inh true
        :requiredPerms 2
        :checkAsUser 10
        :selectedCols (b 8 10)
        :insertedCols (b)
        :updatedCols (b)
      }
      {RTEPERMISSIONINFO
        :relid 18136
        :inh true
        :requiredPerms 2
        :checkAsUser 10
        :selectedCols (b 8 9)
        :insertedCols (b)
        :updatedCols (b)
      }
    )
  )
:jointree
  {FROMEXPR
    :fromlist (
      {RANGETBLREF
        :rtindex 1
      }
      {RANGETBLREF
        :rtindex 2
      }
    )
  }
:quals
  {BOOLEXP
    :boolop and
    :args (
      {OPEXP
        :opno 96
        :opfuncid 65
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
          {VAR
            :varno 1
            :varattno 1
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 1
            :location -1
          }
          {VAR
            :varno 1
            :varattno 4
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 1
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 4
            :location -1
          }
        )
        :location -1
      }
      {OPEXP
        :opno 96
        :opfuncid 65
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
          {VAR

```

```

:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
)
:location -1
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location -1
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
)
:location -1
}
)
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
)
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname s_supkey
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1

```

```

:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname s_name
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 1043
:vartypmod 44
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname s_address
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname s_nationkey
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 1042
:vartypmod 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname s_phone
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname s_acctbal
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 1043
:vartypmod 105
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname s_comment
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 7
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>

```

```

        :stmt_location -1
        :stmt_len -1
    }
}
:security_barrier true
:relid 18291
:inh false
:relkind v
:rellockmode 1
:perminfoindex 4
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTRY
:alias <>
:eref
{ALIAS
:aliasname nation
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:relid 18130
:inh false
:relkind r
:rellockmode 1
:perminfoindex 5
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTRY
:alias <>
:eref
{ALIAS
:aliasname region
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 18136
:inh false
:relkind r
:rellockmode 1
:perminfoindex 6
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTRY
:alias <>
:eref
{ALIAS
:aliasname *GROUP*
:colnames ("n_name")
}
:rtekind 9
:groupexprs (
{VAR
:varno 5
:varattno 2
:varatype 1042
:varitypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 2
:location 51
}
)
:lateral false
:inFromCl false
:securityQuals <>
}
}
:rteperminfos (
{RTEPERMISSIONINFO
:relid 18286
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 18182
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 12)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 18295
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10 13 14)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 18291
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 18130
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 18136
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rindex 1
}
{RANGETBLREF
:rindex 2
}
{RANGETBLREF
:rindex 3
}
{RANGETBLREF
:rindex 4
}
)
}

```

```

{RANGETBLREF
:rtindex 5
}
{RANGETBLREF
:rtindex 6
}
)
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 259
}
{VAR
:varno 2
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location 271
}
)
:location 269
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 293
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 306
}
)
:location 304
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 3
:location 329
}
{VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location 341
}
)
:location 339
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 363
}
{VAR
:varno 4
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0

```

```

:varreturningtype 0
:varnosyn 4
:varattnosyn 4
:location 377
}
)
:location 375
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 4
:location 401
}
{VAR
:varno 5
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location 415
}
)
:location 413
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 5
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 3
:location 439
}
{VAR
:varno 6
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 1
:location 453
}
)
:location 451
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 6
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 2
:location 477
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 486
:constvalue 8 [ 32 0 0 0 65 83 73 65 ]
}
)
:location 484
}
{OPEXPR
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 5
:location 505
}
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 525
:constvalue 4 [ -34 -8 -1 -1 -1 -1 -1 -1 ]
}
)

```

```

:location 517
}
{OPEXPR
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 5
:location 550
}
{CONST
:consttype 1114
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 32 -39 -79 49 -115 -1 -1 ]
}
)
:location 562
}
)
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 5
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 2
:location -1
}
:resno 1
:resname n_name
:ressortgroupref 2
:resorigtbl 18130
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2114
:aggtype 1700
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes (o 1700)
:aggdirectargs <>
:args (
{TARGETENTRY
:expr
{OPEXPR
:opno 1760
:opfuncid 1726
:opresulttype 1700
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 6
:location 71
}
{OPEXPR
:opno 1759
:opfuncid 1725
:opresulttype 1700
:opretset false
:opcollid 0
:inputcollid 0
:args (
{CONST
:consttype 1700
:consttypmod -1
:constcollid 0
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 8 [ 32 0 0 0 0 -128 1 0 ]
}
{VAR
:varno 3
:varattno 7
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 7
:location 94
}
)
:location 92
}
)
:location 87
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar false
:aggvariadic false

```

```

      :aggkind n
      :aggpresorted false
      :agglevelsup 0
      :aggsplit 0
      :aggno -1
      :aggtransno -1
      :location 67
    }
  :resno 2
  :resname revenue
  :ressortgroupref 1
  :resorigtbl 0
  :resorigcol 0
  :resjunk false
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
  {SORTGROUPCLAUSE
    :tleSortGroupRef 2
    :eqop 1054
    :sortop 1058
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
  {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 1752
    :sortop 1756
    :reverse_sort true
    :nulls_first true
    :hashable true
  }
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

```

```

WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1})
{VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
{VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1}) | parallel_safe = true | reason = PROP
ARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:

```

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:etelist <>
:rttable (
  {RANGETBENTRY
    :alias
    {ALIAS
      :aliasname c
      :colnames <>
    }
    :eref
    {ALIAS
      :aliasname c
      :colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone" "
c_acctbal" "c_mktsegment" "c_comment")
    }
    :rtekind 0
    :relid 18171
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
  {RANGETBENTRY
    :alias
    {ALIAS
      :aliasname n
      :colnames <>
    }
    :eref
    {ALIAS
      :aliasname n
      :colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
    }
    :rtekind 0
    :relid 18130
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 2
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
  {RANGETBENTRY
    :alias
    {ALIAS
      :aliasname r
      :colnames <>
    }
    :eref
    {ALIAS
      :aliasname r
      :colnames ("r_regionkey" "r_name" "r_comment")
    }
    :rtekind 0
    :relid 18136
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 3
    :tablesample <>
    :lateral false
  }
}

```

```

:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 18171
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14 15)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 18130
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 18136
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals (
{OPEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{CONST
:consttype 1700
:consttypmod -1
:constcollid 0
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 6 [ 24 0 0 0 0 -128 ]
}
)
:location -1
}
)
}
:rarg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 2
}
{RANGETBLREF
:rtindex 3
}
)
:quals <>
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
)
:location -1
}
)
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 3
:vartype 23
:vartypmod -1

```

```

:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 3
:location -1
}
{VAR
:varno 3
:varattno 1
:varatype 23
:varatypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
}
:location -1
}
{SCALARARRAYOEXPR
:opno 1054
:opfuncid 1048
:hashfuncid 0
:negfuncid 0
:useOr true
:inputcollid 100
:args (
{VAR
:varno 3
:varattno 2
:varatype 1042
:varatypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 2
:location -1
}
{CONST
:consttype 1014
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 48 [ -64 0 0 0 1 0 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0
0 0 40 0 0 0 69 85 82 79 80 69 0 0 44 0 0 0 65 77 69 82 73 67
65 0 ]
}
}
:location -1
}
)
:alias <>
:rtindex 0
}
)
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:varatype 23
:varatypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
}
:resno 1
:resname c_custkey
:ressortgroupref 0
:resorigtbl 18171
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1043
:consttypmod 29
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
}
:resno 2
:resname c_name
:ressortgroupref 0
:resorigtbl 18171
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1043
:consttypmod 44
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
}
:resno 3
:resname c_address
:ressortgroupref 0
:resorigtbl 18171
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:varatype 23
:varatypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
}
:resno 4
:resname c_nationkey

```

```

:ressortgroupref 0
:resorigtbl 18171
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1042
:consttypmod 19
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 5
:resname c_phone
:ressortgroupref 0
:resorigtbl 18171
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1700
:consttypmod 983046
:constcollid 0
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 6
:resname c_acctbal
:ressortgroupref 0
:resorigtbl 18171
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1042
:consttypmod 14
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 7
:resname c_mktsegment
:ressortgroupref 0
:resorigtbl 18171
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1043
:consttypmod 121
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 8
:resname c_comment
:ressortgroupref 0
:resorigtbl 18171
:resorigcol 8
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 1756 :opfuncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 6 :vartype 17 00 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1) {CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 6 [ 24 0 0 0 0 -128 ]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 1756 :opfuncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 6 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1) {CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 6 [ 24 0 0 0 0 -128 ]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) {VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) {VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1}) | parallel_safe = true | reason = PROPPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1) {VAR :varno 2 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 3 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1) {VAR :varno 2 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 3 :location -1}) | parallel_safe = true | reason = PROPPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {SCALARARRAYOEXPR :opno 1054 :opfuncid 1048 :hashfuncid 0 :negfuncid 0 :useOr true :inputcollid 100 :args ((VAR :varno 3 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 2 :location -1) {CONST :consttype 1014 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 48 [ -64 0 0 0 1 0 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0 0 40 0 0 0 69 85 82 79 80 69 0 0 44 0 0 0 65 77 69 82 73 67 65 0 ]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 3) :required_relds (b 3) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 4 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {SCALARARRAYOEXPR :opno 1054 :opfuncid 1048 :hashfuncid 0 :negfuncid 0 :useOr true :inputcollid 100 :args ((VAR :varno 3 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 2 :location -1) {CONST :consttype 1014 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 48 [ -64 0 0 0 1 0 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0 0 40 0 0 0 69 85 82 79 80 69 0 0 44 0 0 0 65 77 69 82 73 67 65 0 ]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 3) :required_relds (b 3) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 4 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 3 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 1 :location -1) {VAR :varno 3 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 2 :location -1}) | parallel_safe = true | reason = PROPPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 2 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 3 :location -1) {VAR :varno 3 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (

```

[illegible]

```

:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTRY
:alias
{ALIAS
:aliasname o
:colnames <>
}
}
:eref
{ALIAS
:aliasname o
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtekind 0
:relid 18182
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTRY
:alias
{ALIAS
:aliasname ps
:colnames <>
}
}
:eref
{ALIAS
:aliasname ps
:colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_supplycost" "ps_comment")
}
:rtekind 0
:relid 18163
:inh false
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
:rteperminfos (
{RTEPERMISSIONINFO
:relid 18194
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 18182
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 18163
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 4
:isNatural false
:larg
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals <>
}
}
:larg
{RANGETBLREF
:rtindex 2
}
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 1
:varatype 23
:varitypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 1
:varatype 23
:varitypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
)
}
:location -1
}
{SCALARARRAYOPEXPR
:opno 1054

```

```

:opfuncid 1048
:hashfuncid 0
:negfuncid 0
:useOr true
:inputcollid 100
:args (
  {VAR
    :varno 2
    :varattno 6
    :vartype 1042
    :vartypmod 19
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 6
    :location -1
  }
  {CONST
    :consttype 1014
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location -1
    :constvalue 48 [ -64 0 0 0 1 0 0 0 0 0 0 0 18 4 0 0 2 0 0 0
1 0 0 0 48 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 7
3 71 72 0 0 ]
  }
)
:location -1
}
)
:alias <>
:rtindex 0
}
:rarg
{RANGETBLREF
:rtindex 3
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 3
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 1
    :location -1
  }
  {VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location -1
  }
)
:location -1
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 3
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 2
    :location -1
  }
  {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
  }
)
:location -1
}
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 3
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 3
    :location -1
  }
  {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location -1
  }
)

```

```

        :constvalue 4 [ 0 0 0 0 0 0 0 ]
    }
}
:location -1
}
)
:alias <>
:rtindex 0
}
)
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname l_orderkey
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull true
:location -1
:constvalue <>
}
:resno 2
:resname l_partkey
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname l_suppkey
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull true
:location -1
:constvalue <>
}
:resno 4
:resname l_linenummer
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1700
:consttypmod 983046
:constcollid 0
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 5
:resname l_quantity
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:varatype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname l_extendedprice
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:varatype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)

```

```

:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname l_discount
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1700
:consttypmod 983046
:constcollid 0
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
}
:resno 8
:resname l_tax
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1042
:consttypmod 5
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
}
:resno 9
:resname l_returnflag
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1042
:consttypmod 5
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
}
:resno 10
:resname l_linestatus
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull true
:location -1
:constvalue <>
}
}
:resno 11
:resname l_shipdate
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull true
:location -1
:constvalue <>
}
}
:resno 12
:resname l_commitdate
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull true
:location -1
:constvalue <>
}
}
:resno 13
:resname l_receiptdate
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1042
:consttypmod 29
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
}
:resno 14
:resname l_shipinstruct
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 14
:resjunk false
}
{TARGETENTRY
:expr

```

```

CONST
consttype 1042
consttypmod 14
constcollid 100
constlen -1
constbyval false
constisnull true
location -1
constvalue <>
}
resno 15
resname 1_shipmode
ressortgroupref 0
resorigtbl 18194
resorigcol 15
resjunk false
}
TARGETENTRY
expr
{
CONST
consttype 1043
consttypmod 48
constcollid 100
constlen -1
constbyval false
constisnull true
location -1
constvalue <>
}
resno 16
resname 1_comment
ressortgroupref 0
resorigtbl 18194
resorigcol 16
resjunk false
}
}
override 0
onConflict <>
returningOldAlias <>
returningNewAlias <>
returningList <>
groupClause <>
groupDistinct false
groupsSets <>
havingQual <>
windowClause <>
distinctClause <>
sortClause <>
limitOffset <>
limitCount <>
limitOption 0
rowMarks <>
setOperations <>
constraintDeps <>
withCheckOptions <>
stmt_location -1
stmt_len -1
}
WARNING: is_parallel_safe= INPUT qual = <>
WARNING: is_parallel_safe= OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe= INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 1 :varattnosyn 1 :location -1}) {VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 1 :varattnosyn 3 :location -1} {VAR :varno 1 :varattno 7 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 1 :varattnosyn 7 :location -1} {VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 1 :varattnosyn 2 :location -1})
WARNING: is_parallel_safe= OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 1 :varattnosyn 1 :location -1}) {VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 1 :varattnosyn 3 :location -1} {VAR :varno 1 :varattno 7 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 1 :varattnosyn 7 :location -1} {VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 1 :varattnosyn 2 :location -1})
WARNING: is_parallel_safe= INPUT qual = ((RESTRICTINFO :clause (SCALARARRAYOPEXPR :opno 1054 :opfncid 1048 :hashfncid 0 :negfncid 0 :useor true :inputcollid 100 :args ((VAR :varno 2 :varattno 6 :vartype 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 2 :varattnosyn 6 :location -1) (CONST :consttype 1014 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 4 [ 0 0 0 0 0 0 0 0 ])) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 3) :required_relds (b 3) :incompatible_relds (b) :outer_relds (b) :left_relds (b 3) :right_relds (b) :orclause <> :rinfo_serial 5 :eval_cost.startup 1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :right_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0)) :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :right_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0)) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe= INPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 2 :varattnosyn 1 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe= INPUT qual = ((RESTRICTINFO :clause (OPEXPR :opno 521 :opfncid 147 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 3 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 3 :varattnosyn 3 :location -1) (CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [ 0 0 0 0 0 0 0 0 ])) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 3) :required_relds (b 3) :incompatible_relds (b) :left_relds (b 3) :right_relds (b) :orclause <> :rinfo_serial 5 :eval_cost.startup 1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0)) :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0)) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe= INPUT qual = ({VAR :varno 3 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 3 :varattnosyn 1 :location -1}) {VAR :varno 3 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 3 :varattnosyn 2 :location -1})
WARNING: is_parallel_safe= OUTPUT qual = ({VAR :varno 3 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 3 :varattnosyn 1 :location -1}) {VAR :varno 3 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 3 :varattnosyn 2 :location -1}) | parallel_safe = true | reason = PROP
WARNING: is_parallel_safe= INPUT qual = ((RESTRICTINFO :clause (OPEXPR :opno 96 :opfncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 1 :varattnosyn 1 :location -1}) {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 2 :varattnosyn 1 :location -1}) :location -1) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b 2) :orclause <> :rinfo_serial 6 :eval_cost.startup 1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 96 :right_hasheqoperator 96)) :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 96 :right_hasheqoperator 96)) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe= OUTPUT qual = ((RESTRICTINFO :clause (OPEXPR :opno 96 :opfncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 1 :varattnosyn 1 :location -1}) {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 2 :varattnosyn 1 :location -1}) :location -1) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 3) :required_relds (b 1 3) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b 3) :orclause <> :rinfo_serial 6 :eval_cost.startup 1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 96 :right_hasheqoperator 96)) :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 96 :right_hasheqoperator 96)) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe= INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 1 :varattnosyn 1 :location -1}) {VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 2 :varattnosyn 2 :location -1}) {VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 3 :varattnosyn 3 :location -1}) {VAR :varno 1 :varattno 7 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 1 :varattnosyn 7 :location -1} {VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 1 :varattnosyn 2 :location -1})
WARNING: is_parallel_safe= OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 1 :varattnosyn 1 :location -1}) {VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 3 :varattnosyn 3 :location -1} {VAR :varno 1 :varattno 7 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 1 :varattnosyn 7 :location -1} {VAR :varno 1 :varattno 2 :vartype 23 :vartyp
```

```
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
```

```

WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 4 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1}
{VAR :varno 4 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 4 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 4 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1}
{VAR :varno 4 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 4 :location -1}) | parallel_safe = true | reason = PROP
ARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname s
:colnames <>
}
}
:eref
{ALIAS
:aliasname s
:colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
}
:rtekind 0
:relid 18153
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias
{ALIAS
:aliasname n
:colnames <>
}
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
}
:rtekind 0
:relid 18130
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias
{ALIAS
:aliasname r
:colnames <>
}
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
}
:rtekind 0
:relid 18136
:inh false
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperinfos (
{RTEPERMISSIONINFO
:relid 18153
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 18130
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 18136
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
):jointree
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
}
:quals (
{OEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR

```

```

:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{CONST
:consttype 1700
:constypmod -1
:constcollid 0
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 6 [ 24 0 0 0 0 -128 ]
}
}
:location -1
}
}
}
:rarg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 2
}
{RANGETBLREF
:rtindex 3
}
)
:quals <>
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
)
:location -1
}
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 3
:location -1
}
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
)
:location -1
}
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 3
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 2
:location -1
}
{CONST
:consttype 1042
:constypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
)
}
)

```

```

        :location -1
    }
}
:alias <>
:rtindex 0
}
)
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname s_suppkey
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1042
:consttypmod 29
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 2
:resname s_name
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1043
:consttypmod 44
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 3
:resname s_address
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname s_nationkey
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1042
:consttypmod 19
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 5
:resname s_phone
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1700
:consttypmod 983046
:constcollid 0
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 6
:resname s_acctbal
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1043
:consttypmod 105
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 7
:resname s_comment
:ressortgroupref 0

```

```
--resorightbl 18153  
--resorigcol 7  
--resjunk false  
  
}  
  
) override 0  
::::::::::::::::::::)  
  
WARNING: is_parallel_safe: INPUT qual = (({RESTRICTINFO :clause {OPEXPR :opno 1756 :ofuncid 1720 :presulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varttno 6 :vartype 17 0} :vartypmod 983846 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varsynsn 1 :varattnosyn 6 :location -1}) (CONST :consttype 1700 :constrtypmod -1 :constcollid 0 :constlen -1 :constnbval false :constisnull true :location -1 :constvaluedata { [ 24 0 0 0 -128 ] } }) :location -1)) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <= :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selcost -1 :outer_selcost -1 :mergeoffamilies (o 426 2097) :left_em (&EQUALENCMEMBER :em_expr {VAR :varno 3 :varttno 2 :vartype 1842 :em_jd_domain {JOINDOMAIN :jd_relds (b 1 2 3)}} :right_em (&EQUALENCMEMBER :em_expr {CONST :consttype 1842 :constrtypmod -1 :constcollid 100 :constlen -1 :constnbval 10 [ 40 0 0 69 85 82 79 80 69 ] } :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 3)})) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize 0 :left_mvfreq -1 :right_mvfreq -1 :left_hashseparator 0 :right_hashseparator 0))  
WARNING: is_parallel_safe: OUTPUT qual = (({RESTRICTINFO :clause {OPEXPR :opno 1756 :ofuncid 1720 :presulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varttno 6 :vartype 1 700 :vartypmod 983846 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varsynsn 1 :varattnosyn 6 :location -1}) (CONST :consttype 1700 :constrtypmod -1 :constcollid 0 :constlen -1 :constnbval false :constisnull true :location -1 :constvaluedata { [ 24 0 0 0 -128 ] } }) :location -1)) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <= :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selcost -1 :outer_selcost -1 :mergeoffamilies (o 426 2097) :left_em (&EQUALENCMEMBER :em_expr {VAR :varno 3 :varttno 2 :vartype 1842 :em_jd_domain {JOINDOMAIN :jd_relds (b 1 2 3)}} :right_em (&EQUALENCMEMBER :em_expr {CONST :consttype 1842 :constrtypmod -1 :constcollid 100 :constlen -1 :constnbval 10 [ 40 0 0 69 85 82 79 80 69 ] } :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 3)})) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize 0 :left_mvfreq -1 :right_mvfreq -1 :left_hashseparator 0 :right_hashseparator 0)) | parallel_safe = true | reason = PROP PARALLEL_SAFE  
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varttno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varsynsn 1 :varattnosyn 1 :location -1}) (VAR :varno 1 :varttno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varsynsn 1 :varattnosyn 4 :location -1}))  
WARNING: is_parallel_safe: OUTPUT qual = (({VAR :varno 1 :varttno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varsynsn 1 :varattnosyn 1 :location -1}) (VAR :varno 1 :varttno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varsynsn 1 :varattnosyn 4 :location -1})) | parallel_safe = true | reason = PROP PARALLEL_SAFE  
ARALLEL_SAFE  
WARNING: is_parallel_safe: INPUT qual = <>  
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE  
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 2 :varttno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varsynsn 2 :varattnosyn 1 :location -1}) (VAR :varno 2 :varttno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varsynsn 2 :varattnosyn 3 :location -1}))  
WARNING: is_parallel_safe: OUTPUT qual = (({VAR :varno 2 :varttno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varsynsn 2 :varattnosyn 1 :location -1}) (VAR :varno 2 :varttno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varsynsn 2 :varattnosyn 3 :location -1})) | parallel_safe = true | reason = PROP PARALLEL_SAFE  
ARALLEL_SAFE  
WARNING: is_parallel_safe: INPUT qual = (({RESTRICTINFO :clause {OPEXPR :opno 1854 :ofuncid 1848 :presulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args {(VAR :varno 3 :varttno 2 :vartype 1842 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varsynsn 3 :varattnosyn 2 :location -1}) (CONST :consttype 1842 :constrtypmod -1 :constcollid 100 :constlen -1 :constnbval false :constisnull true :location -1 :constvaluedata { [ 40 0 0 69 85 82 79 80 69 ] } }) :location -1)) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 3) :required_relds (b 3) :incompatible_relds (b) :outer_relds (b) :left_relds (b 3) :right_relds (b) :orclause <= :rinfo_serial 4 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selcost -1 :outer_selcost -1 :mergeoffamilies (o 426 2097) :left_em (&EQUALENCMEMBER :em_expr {VAR :varno 3 :varttno 2 :vartype 1842 :em_jd_domain {JOINDOMAIN :jd_relds (b 1 2 3)}} :right_em (&EQUALENCMEMBER :em_expr {CONST :consttype 1842 :constrtypmod -1 :constcollid 100 :constlen -1 :constnbval 10 [ 40 0 0 69 85 82 79 80 69 ] } :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 3)})) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvfreq -1 :right_mvfreq -1 :left_hashseparator 0 :right_hashseparator 0))  
WARNING: is_parallel_safe: OUTPUT qual = (({RESTRICTINFO :clause {OPEXPR :opno 1854 :ofuncid 1848 :presulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args {(VAR :varno 3 :varttno 2 :vartype 1842 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varsynsn 3 :varattnosyn 2 :location -1}) (CONST :consttype 1842 :constrtypmod -1 :constcollid 100 :constlen -1 :constnbval false :constisnull true :location -1 :constvaluedata { [ 40 0 0 69 85 82 79 80 69 ] } }) :location -1)) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 3) :required_relds (b 3) :incompatible_relds (b) :outer_relds (b) :left_relds (b 3) :right_relds (b) :orclause <= :rinfo_serial 4 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selcost -1 :outer_selcost -1 :mergeoffamilies (o 426 2097) :left_em (&EQUALENCMEMBER :em_expr {VAR :varno 3 :varttno 2 :vartype 1842 :em_jd_domain {JOINDOMAIN :jd_relds (b 1 2 3)}} :right_em (&EQUALENCMEMBER :em_expr {CONST :consttype 1842 :constrtypmod -1 :constcollid 100 :constlen -1 :constnbval 10 [ 40 0 0 69 85 82 79 80 69 ] } :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 3)})) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvfreq -1 :right_mvfreq -1 :left_hashseparator 0 :right_hashseparator 0)) | parallel_safe = true | reason = PROPARALLEL_SAFE  
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 3 :varttno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varsynsn 3 :varattnosyn 1 :location -1}) (VAR :varno 3 :varttno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varsynsn 3 :varattnosyn 2 :location -1})) | parallel_safe = true | reason = PROPARALLEL_SAFE  
WARNING: is_parallel_safe: OUTPUT qual = (({RESTRICTINFO :clause {OPEXPR :opno 96 :ofuncid 65 :presulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 2 :varttno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varsynsn 2 :varattnosyn 3 :location -1}) (VAR :varno 3 :varttno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varsynsn 3 :varattnosyn 1 :location -1})) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 2 3) :required_relds (b 2 3) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b 3) :orclause <= :rinfo_serial 5 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selcost -1 :outer_selcost -1 :mergeoffamilies (o 1976) :left_em (&EQUALENCMEMBER :em_expr {VAR :varno 2 :varttno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varsynsn 2 :varattnosyn 3 :location -1}) :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 3)}} :right_em (&EQUALENCMEMBER :em_expr {VAR :varno 3 :varttno 1 :v
```

[illegible]

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1:var_tynpmod -1:var_collid 0:varnullingrels (b) :varlevelsup 0:varreturningtype 0:varnosyn 5:varattnosyn 3:location -1:em_relds (b 5) :em_is_const false :em_is_child false :em_data_type 23 :va
2:var_collid (JOINDOMAIN :jd_relds (b 1 2 3 4 5 6)) :right_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 4:varattno 1:var_tynp 23:var_tynpmod -1:var_collid 0:varnullingrels (b) :varlevelsup 0:varreturning
n_tynp 0:varnosyn 6:varattnosyn 1:location -1) :em_relds (b 6) :em_is_const false :em_is_child false :em_data_type 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2 3 4 5 6)) :outer_is_left false :hashjoino
operator 96 :left_bucketsize 0:0658823529411764705 :right_bucketsize 1:left_mcvfreq 0:right_mcvfreq 0:left_hashoperator 96:right_hashoperator 96) | parallel_safe = true | reason = PROPAGATE_SAFE
_WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 1:varattno 1:var_tynp 23:var_tynpmod -1:var_collid 0:varnullingrels (b) :varlevelsup 0:varreturningtype 0:varnosyn 4:varattnosyn 1:location -1)
(VAR :varno 4:varattno 4:var_tynp 23:var_tynpmod -1:var_collid 0:varnullingrels (b) :varlevelsup 0:varreturningtype 0:varnosyn 4:varattnosyn 2:location -1) (VAR :varno 5:varattno 2:var_tynp 1842:va
r_tynpmod -1:var_collid 100:varnullingrels (b) :varlevelsup 0:varreturningtype 0:varnosyn 5:varattnosyn 2:location -1) (VAR :varno 5:varattno 1:var_tynp 23:var_tynpmod -1:var_collid 0:varnullingr
els (b) :varlevelsup 0:varreturningtype 0:varnosyn 5:varattnosyn 1:location -1)
_WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 4:varattno 1:var_tynp 23:var_tynpmod -1:var_collid 0:varnullingrels (b) :varlevelsup 0:varreturningtype 0:varnosyn 4:varattnosyn 1:location -1)
(VAR :varno 4:varattno 4:var_tynp 23:var_tynpmod -1:var_collid 0:varnullingrels (b) :varlevelsup 0:varreturningtype 0:varnosyn 4:varattnosyn 2:location -1) (VAR :varno 5:varattno 2:var_tynp 1842:va
r_tynpmod -1:var_collid 100:varnullingrels (b) :varlevelsup 0:varreturningtype 0:varnosyn 5:varattnosyn 2:location -1) (VAR :varno 5:varattno 1:var_tynp 23:var_tynpmod -1:var_collid 0:varnullingr
els (b) :varlevelsup 0:varreturningtype 0:varnosyn 5:varattnosyn 1:location -1)
_WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 5:varattno 2:var_tynp 1842:var_tynpmod -1:var_collid 100:varnullingrels (b) :varlevelsup 0:varreturningtype 0:varnosyn 5:varattnosyn 2:location -1)
(VAR :varno 5:varattno 1:var_tynp 23:var_tynpmod -1:var_collid 0:varnullingrels (b) :varlevelsup 0:varreturningtype 0:varnosyn 5:varattnosyn 1:location -1)
_WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 5:varattno 2:var_tynp 1842:var_tynpmod -1:var_collid 100:varnullingrels (b) :varlevelsup 0:varreturningtype 0:varnosyn 5:varattnosyn 2:location -1)
(VAR :varno 5:varattno 1:var_tynp 23:var_tynpmod -1:var_collid 0:varnullingrels (b) :varlevelsup 0:varreturningtype 0:varnosyn 5:varattnosyn 1:location -1)
_WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause (OPEXPR :opno 96:opfuncid 65:opresulttype 16:opretset false:opcollid 0:inputcollid 0:args ((VAR :varno 3:varattno 3:var_tynp 23:va
r_tynpmod -1:var_collid 0:varnullingrels (b) :varlevelsup 0:varreturningtype 0:varnosyn 3:varattnosyn 3:location -1) (VAR :varno 4:varattno 1:var_tynp 23:var_tynpmod -1:var_collid 0:varnullingrels (
b) :varlevelsup 0:varreturningtype 0:varnosyn 4:varattnosyn 1:location -1)) :location -1) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false
:has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 3 4) :required_relds (b 3 4) :incompatible_relds (b) :outer_relds (b 3) :right_relds (b 4) :orclause <> :rinfo_s
erial 12 :eval_cost.startup 0 :eval_cost.per_tuple 0.0025 :norm_selc 0.005 :outer_selc 1 :mergeofamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 1:varattno 4:var_tynp 23:var_tynpmod
-1:var_collid 0:varnullingrels (b) :varlevelsup 0:varreturningtype 0:varnosyn 3:varattnosyn 3:location -1) :em_relds (b 3) :em_is_const false :em_is_child false :em_data_type 23 :em_jdomain (JOI
N DOMAIN :jd_relds (b 1 2 3 4 5 6))) :right_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 1:varattno 4:var_tynp 23:var_tynpmod -1:var_collid 0:varnullingrels (b) :varlevelsup 0:varreturningtype 0:varnosyn
n 4:varattnosyn 1:location -1) :em_relds (b 4) :em_is_const false :em_is_child false :em_data_type 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2 3 4 5 6))) :outer_is_left true :hashjoinoperator 96 :lef
t_bucketsize 0.1 :right_bucketsize 0.0999999999999991 :left_mcvfreq 0:right_mcvfreq 0:left_hashoperator 96:right_hashoperator 96) (RESTRICTINFO :clause (OPEXPR :opno 96:opfuncid 65:opresulttype
16:opretset false:opcollid 0:inputcollid 0:args ((VAR :varno 1:varattno 4:var_tynp 23:var_tynpmod -1:var_collid 0:varnullingrels (b) :varlevelsup 0:varreturningtype 0:varnosyn 4:varattnosyn 4:locat
ion -1) (VAR :varno 4:varattno 4:var_tynp 23:var_tynpmod -1:var_collid 0:varnullingrels (b) :varlevelsup 0:varreturningtype 0:varnosyn 4:varattnosyn 4:location -1)) :location -1) :is_pushed_
down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 4) :required_relds (b 1 4) :incom
patible_relds (b) :outer_relds (b 1) :right_relds (b 4) :orclause <> :rinfo_serial 11 :eval_cost.startup 0 :eval_cost.per_tuple 0.0025 :norm_selc 0.005 :outer_selc 1 :mergeofamilies (o 1976) :left_em
(EQUIVALENCEMEMBER :em_expr (VAR :varno 1:varattno 4:var_tynp 23:var_tynpmod -1:var_collid 0:varnullingrels (b) :varlevelsup 0:varreturningtype 0:varnosyn 4:varattnosyn 4:location -1) :em_relds (b 1) :em_is_const false :em_is_child false :em_data_type 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2 3 4 5 6))) :right_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 4:varattno 4:var_tynp
23:var_tynpmod -1:var_collid 0:varnullingrels (b) :varlevelsup 0:varreturningtype 0:varnosyn 4:varattnosyn 4:location -1) :em_relds (b 4) :em_is_const false :em_is_child false :em_data_type 23 :e
m_jdomain (JOINDOMAIN :jd_relds (b 1 2 3 4 5 6))) :outer_is_left true :hashjoinoperator 96 :left_bucketsize 0.1 :right_bucketsize 0.0999999999999991 :left_mcvfreq 0:right_mcvfreq 0:left_hashoperator
96:right_hashoperator 96) | parallel_safe = true | reason = PROPAGATE_SAFE
_WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause (OPEXPR :opno 96:opfuncid 65:opresulttype 16:opretset false:opcollid 0:inputcollid 0:args ((VAR :varno 3:varattno 3:var_tynp 23:va
r_tynpmod -1:var_collid 0:varnullingrels (b) :varlevelsup 0:varreturningtype 0:varnosyn 3:varattnosyn 3:location -1) (VAR :varno 4:varattno 1:var_tynp 23:var_tynpmod -1:var_collid 0:varnullingrels (
b) :varlevelsup 0:varreturningtype 0:varnosyn 4:varattnosyn 1:location -1)) :location -1) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false
:has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 3 4) :required_relds (b 3 4) :incompatible_relds (b) :outer_relds (b 3) :right_relds (b 4) :orclause <> :rinfo_s
erial 14 :eval_cost.startup 0 :eval_cost.per_tuple 0.0025 :norm_selc 0.005 :outer_selc 1 :mergeofamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 3:varattno 3:var_tynp 23:var_tynpmod
-1:var_collid 0:varnullingrels (b) :varlevelsup 0:varreturningtype 0:varnosyn 3:varattnosyn 3:location -1) :em_relds (b 3) :em_is_const false :em_is_child false :em_data_type 23 :em_jdomain (JOI
N DOMAIN :jd_relds (b 1 2 3 4 5 6))) :right_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 1:varattno 4:var_tynp 23:var_tynpmod -1:var_collid 0:varnullingrels (b) :varlevelsup 0:varreturningtype 0:varnosyn
n 4:varattnosyn 1:location -1) :em_relds (b 4) :em_is_const false :em_is_child false :em_data_type 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2 3 4 5 6))) :outer_is_left true :hashjoinoperator 96 :lef
t_bucketsize 0.1 :right_bucketsize 0.0999999999999991 :left_mcvfreq 0:right_mcvfreq 0:left_hashoperator 96:right_hashoperator 96) (RESTRICTINFO :clause (OPEXPR :opno 96:opfuncid 65:opresulttype
16:opretset false:opcollid 0:inputcollid 0:args ((VAR :varno 1:varattno 4:var_tynp 23:var_tynpmod -1:var_collid 0:varnullingrels (b) :varlevelsup 0:varreturningtype 0:varnosyn 1:varattnosyn 4:locat
ion -1) (VAR :varno 4:varattno 4:var_tynp 23:var_tynpmod -1:var_collid 0:varnullingrels (b) :varlevelsup 0:varreturningtype 0:varnosyn 4:varattnosyn 4:location -1)) :location -1) :is_pushed_
down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 4) :required_relds (b 1 4) :incom
patible_relds (b) :outer_relds (b 1) :right_relds (b 4) :orclause <> :rinfo_serial 11 :eval_cost.startup 0 :eval_cost.per_tuple 0.0025 :norm_selc 0.005 :outer_selc 1 :mergeofamilies (o 1976) :left_em
(EQUIVALENCEMEMBER :em_expr (VAR :varno 1:varattno 4:var_tynp 23:var_tynpmod -1:var_collid 0:varnullingrels (b) :varlevelsup 0:varreturningtype 0:varnosyn 4:varattnosyn 4:location -1) :em_relds (b 1) :em_is_const false :em_is_child false :em_data_type 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2 3 4 5 6))) :right_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 4:varattno 4:var_tynp
23:var_tynpmod -1:var_collid 0:varnullingrels (b) :varlevelsup 0:varreturningtype 0:varnosyn 4:varattnosyn 4:location -
```

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QUERY PLAN

QUERY PLAN

```
-> Hash Join (cost=12939.67..50937.23 rows=11058 width=20) (actual time=78.189..270.038 rows=27362.33 loops=3)
  Output: l1_extendedprice, l1_discount, l1_orderkey, supplier_secure.s_nationkey
  Hash Cond: (l1_suppkey = supplier_secure.s_suppkey)
  Buffers: shared hit=23 read=31086
  Worker 0:  actual time=78.237..269.973 rows=27374.00 loops=1
    Buffers: shared hit=125 read=18454
  Worker 1:  actual time=78.234..269.369 rows=27274.00 loops=1
    Buffers: shared hit=146 read=18312
-> Parallel Hash Semi Join (cost=12848.07..47962.45 rows=201062 width=346) (actual time=77.043..253.952 rows=160171.00 loops=3)
  Output: l1_orderkey, NULL::integer, l1_suppkey, NULL::integer, NULL::numeric(15,2), l1_extendedprice, l1_discount, NULL::numeric(15,2), NULL::character(1), NULL::character(1), NULL::date, NULL::date, NULL::character(25), NULL::character(10), NULL::character varying(44)
  Hash Cond: ((l1_partkey = ps_ps_partkey) AND (l1_suppkey = ps_ps_suppkey))
  Buffers: shared hit=282 read=31086
  Worker 0:  actual time=77.094..253.633 rows=160930.00 loops=1
    Buffers: shared hit=78 read=10454
  Worker 1:  actual time=77.103..253.296 rows=159154.00 loops=1
    Buffers: shared hit=99 read=10312
-> Parallel Hash Semi Join (cost=7462.84..38563.98 rows=201062 width=24) (actual time=25.505..131.582 rows=160171.00 loops=3)
  Output: l1_orderkey, l1_suppkey, l1_extendedprice, l1_discount, l1_partkey
  Hash Cond: (l1_orderkey = o.o_orderkey)
  Buffers: shared hit=282 read=27542
  Worker 0:  actual time=25.552..131.063 rows=160930.00 loops=1
    Buffers: shared hit=78 read=9252
  Worker 1:  actual time=25.568..131.443 rows=159154.00 loops=1
    Buffers: shared hit=99 read=9128
-> Parallel Seq Scan on public.lineitem l (cost=0.00..27551.87 rows=499987 width=24) (actual time=0.415..28.841 rows=399989.67 loops=3)
  Output: l1_orderkey, l1_partkey, l1_partkey, l1_linenum, l1_linenum, l1_quantity, l1_extendedprice, l1_discount, l1_tax, l1_returnflag, l1_linestatus, l1_shipdate, l1_commitdate, l1_receiptdate, l1_shipinstruct, l1_shipmode, l1_comment
  Buffers: shared read=22552
  Worker 0:  actual time=0.469..28.215 rows=402470.00 loops=1
    Buffers: shared read=7566
  Worker 1:  actual time=0.466..28.968 rows=398105.00 loops=1
    Buffers: shared read=7482
-> Parallel Hash (cost=6834.50..6834.50 rows=50267 width=4) (actual time=24.997..24.997 rows=40048.00 loops=3)
  Output: o.o_orderkey
  Buckets: 131072 Batches: 1 Memory Usage: 5760kB
  Buffers: shared hit=282 read=4990
  Worker 0:  actual time=25.056..25.056 rows=40074.00 loops=1
    Buffers: shared hit=78 read=1686
  Worker 1:  actual time=25.060..25.060 rows=39906.00 loops=1
    Buffers: shared hit=99 read=1646
-> Parallel Seq Scan on public.orders o (cost=0.00..6834.50 rows=50267 width=4) (actual time=0.038..19.574 rows=40048.00 loops=3)
  Output: o.o_orderkey
  Filter: (o.o_orderpriority = ANY ('{1-URGENT,2-HIGH}'))::bpchar[])
  Rows Removed by Filter: 59952
  Buffers: shared hit=282 read=4990
  Worker 0:  actual time=0.048..19.708 rows=40074.00 loops=1
    Buffers: shared hit=78 read=1686
  Worker 1:  actual time=0.032..19.374 rows=39906.00 loops=1
    Buffers: shared hit=99 read=1646
-> Parallel Hash (cost=4377.33..4377.33 rows=66660 width=8) (actual time=51.366..51.367 rows=53333.33 loops=3)
  Output: ps_ps_partkey, ps_ps_suppkey
  Buckets: 262144 Batches: 1 Memory Usage: 8352kB
  Buffers: shared read=3544
  Worker 0:  actual time=51.065..51.065 rows=54564.00 loops=1
    Buffers: shared read=1202
  Worker 1:  actual time=51.520..51.520 rows=52815.00 loops=1
    Buffers: shared read=1184
-> Parallel Seq Scan on public.partsupp ps (cost=0.00..4377.33 rows=66660 width=8) (actual time=0.513..42.173 rows=53333.33 loops=3)
  Output: ps_ps_partkey, ps_ps_suppkey
  Filter: (ps_ps_availqty > 0)
  Buffers: shared read=3544
  Worker 0:  actual time=0.429..42.141 rows=54564.00 loops=1
    Buffers: shared read=1202
  Worker 1:  actual time=0.713..42.170 rows=52815.00 loops=1
    Buffers: shared read=1184
-> Hash (cost=99.46..99.46 rows=11 width=8) (actual time=1.107..1.109 rows=342.00 loops=3)
  Output: supplier_secure.s_suppkey, supplier_secure.s_nationkey
  Buckets: 1024 Batches: 1 Memory Usage: 22kB
  Buffers: shared hit=141
  Worker 0:  actual time=1.117..1.118 rows=342.00 loops=1
    Buffers: shared hit=47
  Worker 1:  actual time=1.095..1.097 rows=342.00 loops=1
    Buffers: shared hit=47
-> Subquery Scan on supplier_secure (cost=24.50..99.46 rows=11 width=8) (actual time=0.179..1.065 rows=342.00 loops=3)
  Output: supplier_secure.s_suppkey, supplier_secure.s_nationkey
  Buffers: shared hit=141
  Worker 0:  actual time=0.210..1.075 rows=342.00 loops=1
    Buffers: shared hit=47
  Worker 1:  actual time=0.230..1.052 rows=342.00 loops=1
    Buffers: shared hit=47
-> Hash Semi Join (cost=24.50..99.35 rows=11 width=512) (actual time=0.177..1.038 rows=342.00 loops=3)
  Output: s.s_suppkey, NULL::character(25), NULL::character varying(40), s.s_nationkey, NULL::character(15), NULL::numeric(15,2), NULL::character varying(101)
  Hash Cond: (s.s_nationkey = n.n_nationkey)
  Buffers: shared hit=141
  Worker 0:  actual time=0.209..1.048 rows=342.00 loops=1
    Buffers: shared hit=47
  Worker 1:  actual time=0.227..1.024 rows=342.00 loops=1
    Buffers: shared hit=47
-> Seq Scan on public.supplier s (cost=0.00..70.00 rows=1801 width=8) (actual time=0.064..0.795 rows=1802.00 loops=3)
  Output: s.s_suppkey, s.s_name, s.s_address, s.s_nationkey, s.s_phone, s.s_acctbal, s.s_comment
  Filter: (s.s_acctbal > '0')::numeric)
  Rows Removed by Filter: 198
  Buffers: shared hit=135
  Worker 0:  actual time=0.095..0.807 rows=1802.00 loops=1
    Buffers: shared hit=45
  Worker 1:  actual time=0.077..0.747 rows=1802.00 loops=1
    Buffers: shared hit=45
-> Hash (cost=24.48..24.48 rows=1 width=4) (actual time=0.103..0.104 rows=5.00 loops=3)
  Output: n.n_nationkey
  Buckets: 1024 Batches: 1 Memory Usage: 9kB
  Buffers: shared hit=6
  Worker 0:  actual time=0.105..0.105 rows=5.00 loops=1
    Buffers: shared hit=2
  Worker 1:  actual time=0.139..0.140 rows=5.00 loops=1
    Buffers: shared hit=2
-> Hash Join (cost=12.14..24.48 rows=1 width=4) (actual time=0.095..0.098 rows=5.00 loops=3)
  Output: n.n_nationkey
  Hash Cond: (n.n_nationkey = r.r_regionkey)
  Buffers: shared hit=6
  Worker 0:  actual time=0.096..0.100 rows=5.00 loops=1
    Buffers: shared hit=2
  Worker 1:  actual time=0.131..0.134 rows=5.00 loops=1
    Buffers: shared hit=2
-> Seq Scan on public.nation n (cost=0.00..11.70 rows=170 width=8) (actual time=0.041..0.042 rows=25.00 loops=3)
  Output: n.n_nationkey, n.n_name, n.n_regionkey, n.n_comment
  Buffers: shared hit=3
  Worker 0:  actual time=0.041..0.042 rows=25.00 loops=1
    Buffers: shared hit=1
  Worker 1:  actual time=0.064..0.066 rows=25.00 loops=1
    Buffers: shared hit=1
-> Hash (cost=12.12..12.12 rows=1 width=4) (actual time=0.043..0.043 rows=1.00 loops=3)
  Output: r.r_regionkey
  Buckets: 1024 Batches: 1 Memory Usage: 9kB
  Buffers: shared hit=3
  Worker 0:  actual time=0.046..0.047 rows=1.00 loops=1
    Buffers: shared hit=1
  Worker 1:  actual time=0.055..0.055 rows=1.00 loops=1
    Buffers: shared hit=1
-> Seq Scan on public.region r (cost=0.00..12.12 rows=1 width=4) (actual time=0.036..0.037 rows=1.00 loops=3)
  Output: r.r_regionkey
  Filter: (r.r_name = 'EUROPE')::bpchar)
  Rows Removed by Filter: 4
  Buffers: shared hit=3
  Worker 0:  actual time=0.040..0.041 rows=1.00 loops=1
    Buffers: shared hit=1
  Worker 1:  actual time=0.049..0.049 rows=1.00 loops=1
    Buffers: shared hit=1
-> Parallel Hash (cost=7147.00..7147.00 rows=19220 width=8) (actual time=60.665..60.665 rows=15280.33 loops=3)
  Output: orders.o_custkey, orders.o_orderkey
  Buckets: 65536 Batches: 1 Memory Usage: 2336kB
  Buffers: shared hit=94 read=5178
  Worker 0:  actual time=56.276..56.277 rows=12120.00 loops=1
    Buffers: shared hit=49 read=1368
  Worker 1:  actual time=56.272..56.272 rows=11642.00 loops=1
    Buffers: shared hit=9 read=1317
-> Parallel Seq Scan on public.orders (cost=0.00..7147.00 rows=19220 width=8) (actual time=0.892..57.708 rows=15280.33 loops=3)
```

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Output: orders.o_custkey, orders.o_orderkey
Filter: ((orders.o_orderdate >= '1996-01-01'::date) AND (orders.o_orderdate < '1996-01-01 00:00:00'::timestamp without time zone))
Rows Removed by Filter: 84728
Buffers: shared hit=94 read=5178
Worker 0: actual time=1.169..54.142 rows=12120.00 loops=1
Buffers: shared hit=49 read=1368
Worker 1: actual time=1.203..53.758 rows=11642.00 loops=1
Buffers: shared hit=9 read=1317
-> Materialize (cost=24.50..24.50 rows=1 width=108) (actual time=0.068..0.070 rows=5.00 loops=3)
Output: nation.n_name, nation.n_nationkey
Storage: Memory Maximum Storage: 17kB
Buffers: shared hit=6
Worker 0: actual time=0.048..0.049 rows=5.00 loops=1
Buffers: shared hit=2
Worker 1: actual time=0.099..0.100 rows=5.00 loops=1
Buffers: shared hit=2
-> Sort (cost=24.50..24.50 rows=1 width=108) (actual time=0.059..0.059 rows=5.00 loops=3)
Output: nation.n_name, nation.n_nationkey
Sort Key: nation.n_nationkey
Sort Method: quicksort Memory: 25kB
Buffers: shared hit=6
Worker 0: actual time=0.040..0.041 rows=5.00 loops=1
Sort Method: quicksort Memory: 25kB
Buffers: shared hit=2
Worker 1: actual time=0.089..0.089 rows=5.00 loops=1
Sort Method: quicksort Memory: 25kB
Buffers: shared hit=2
-> Hash Join (cost=12.14..24.48 rows=1 width=108) (actual time=0.040..0.043 rows=5.00 loops=3)
Output: nation.n_name, nation.n_nationkey
Hash Cond: (nation.n_regionkey = region.r_regionkey)
Buffers: shared hit=6
Worker 0: actual time=0.025..0.028 rows=5.00 loops=1
Buffers: shared hit=2
Worker 1: actual time=0.069..0.072 rows=5.00 loops=1
Buffers: shared hit=2
-> Seq Scan on public.nation (cost=0.00..11.70 rows=170 width=112) (actual time=0.008..0.009 rows=25.00 loops=3)
Output: nation.n_nationkey, nation.n_name, nation.n_regionkey, nation.n_comment
Buffers: shared hit=3
Worker 0: actual time=0.006..0.007 rows=25.00 loops=1
Buffers: shared hit=1
Worker 1: actual time=0.011..0.012 rows=25.00 loops=1
Buffers: shared hit=1
-> Hash (cost=12.12..12.12 rows=1 width=4) (actual time=0.015..0.016 rows=1.00 loops=3)
Output: region.r_regionkey
Buckets: 1024 Batches: 1 Memory Usage: 9kB
Buffers: shared hit=3
Worker 0: actual time=0.008..0.008 rows=1.00 loops=1
Buffers: shared hit=1
Worker 1: actual time=0.028..0.028 rows=1.00 loops=1
Buffers: shared hit=1
-> Seq Scan on public.region (cost=0.00..12.12 rows=1 width=4) (actual time=0.012..0.013 rows=1.00 loops=3)
Output: region.r_regionkey
Filter: (region.r_name = 'ASIA'::bpchar)
Rows Removed by Filter: 4
Buffers: shared hit=3
Worker 0: actual time=0.006..0.006 rows=1.00 loops=1
Buffers: shared hit=1
Worker 1: actual time=0.024..0.024 rows=1.00 loops=1
Buffers: shared hit=1
-> Materialize (cost=24.53..1199.30 rows=320 width=8) (never executed)
Output: customer_secure.c_custkey, customer_secure.c_nationkey
-> Subquery Scan on customer_secure (cost=24.53..1197.70 rows=320 width=8) (never executed)
Output: customer_secure.c_custkey, customer_secure.c_nationkey
-> Hash Semi Join (cost=24.53..1194.50 rows=320 width=52) (never executed)
Output: c.c_custkey, NULL::character varying(25), NULL::character varying(40), c.c_nationkey, NULL::character(15), NULL::numeric(15,2), NULL::character(10), NULL::charac
ter varying(117)
Hash Cond: (c.c_nationkey = n_1.n_nationkey)
-> Seq Scan on public.customer c (cost=0.00..1095.00 rows=27203 width=8) (never executed)
Output: c.c_custkey, c.c_name, c.c_address, c.c_nationkey, c.c_phone, c.c_acctbal, c.c_mktsegment, c.c_comment
Filter: (c.c_acctbal > '0'::numeric)
-> Hash (cost=24.51..24.51 rows=2 width=4) (never executed)
Output: n_1.n_nationkey
-> Hash Join (cost=12.15..24.51 rows=2 width=4) (never executed)
Output: n_1.n_nationkey
Hash Cond: (n_1.n_regionkey = r_1.r_regionkey)
-> Seq Scan on public.nation n_1 (cost=0.00..11.70 rows=170 width=8) (never executed)
Output: n_1.n_nationkey, n_1.n_name, n_1.n_regionkey, n_1.n_comment
-> Hash (cost=12.12..12.12 rows=2 width=4) (never executed)
Output: r_1.r_regionkey
-> Seq Scan on public.region r_1 (cost=0.00..12.12 rows=2 width=4) (never executed)
Output: r_1.r_regionkey
Filter: (r_1.r_name = ANY ('{EUROPE,AMERICA}'::bpchar[]))

Planning:
Buffers: shared hit=292 read=6
Planning Time: 20.119 ms
Execution Time: 351.339 ms
(286 rows)

tpch-200mb-2=>

```